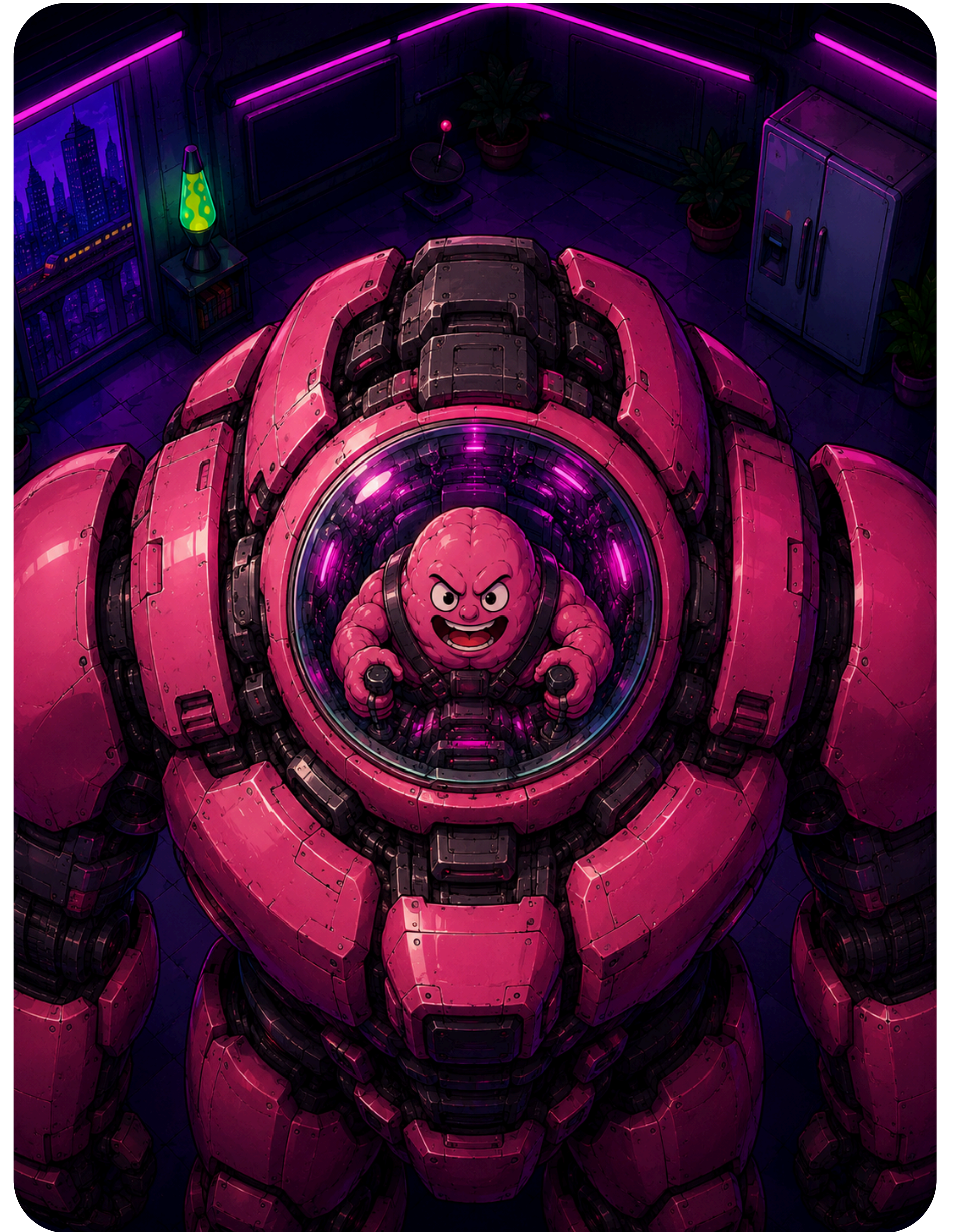


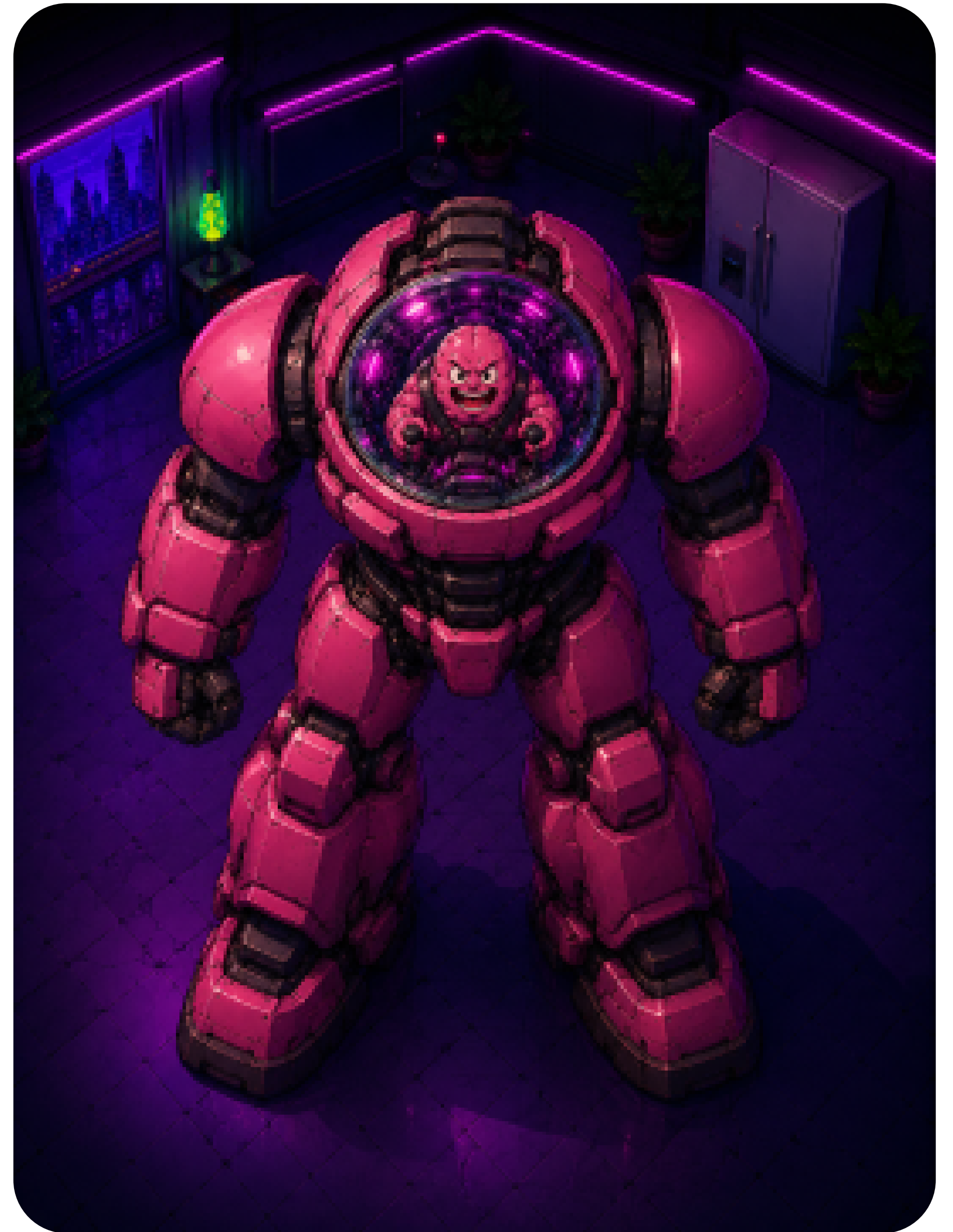
Beneath the Skull of AI: Giving the Brain a Body (Part 2)

By: Oussama Bounouch



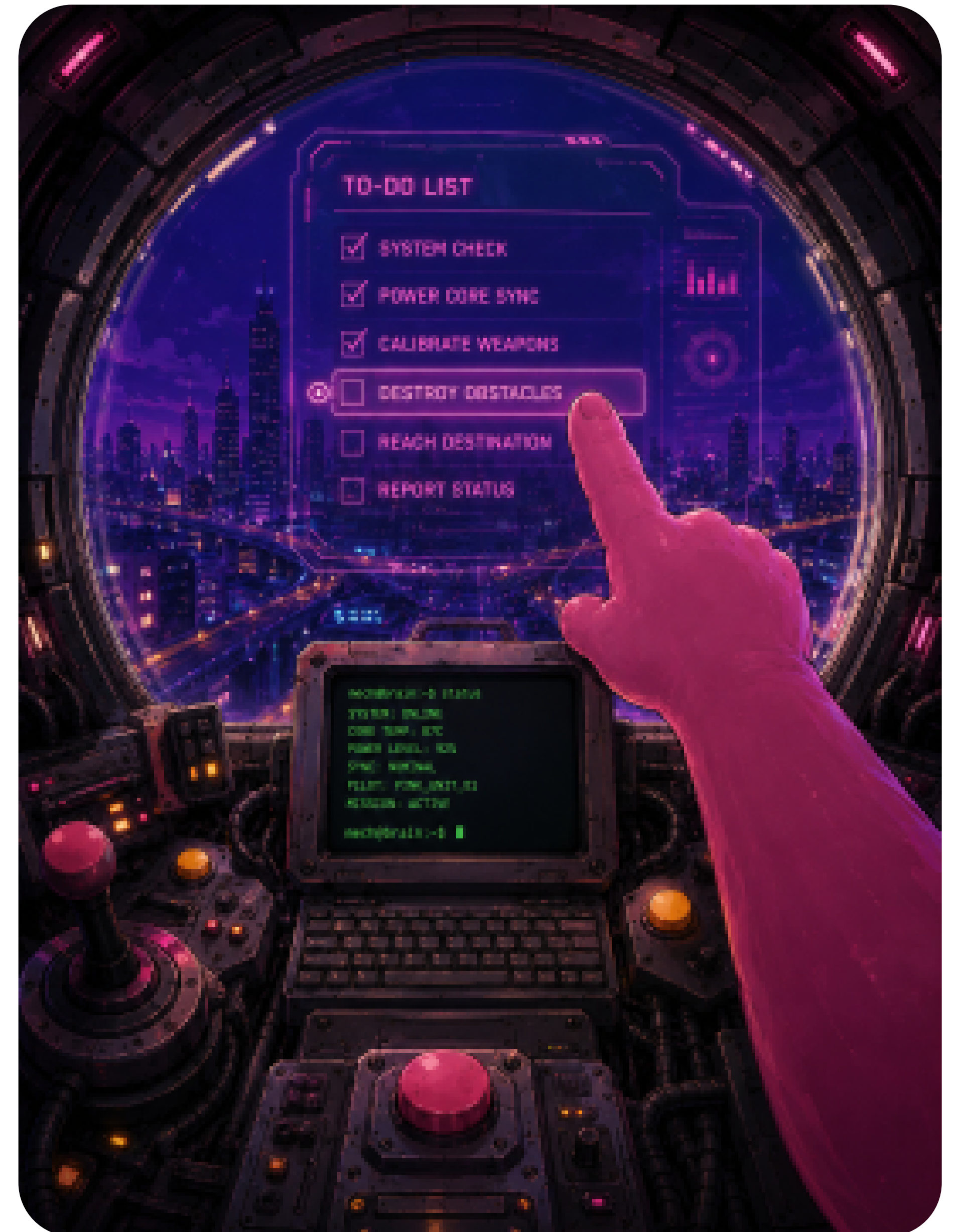
Quick Recap of Part 1:

- The **model (LLM)** is a big file of numbers.
- The **engine** is the software that runs it.
- Your **RAM** decides which model fits.
- **Quantization** Makes large models run on smaller computers.
- The Importance of **Bandwidth** for Calculating Token Speed.
- **Context** compression & quantization...



What You'll Learn Today

- Today, we'll transform an **LLM** into a real **AI Agent**.
- How **agents** see, think, and use tools
- How Hermes is built, its **brain, skills, tools, memory**, and **instructions**.
- How to get started with **Hermes Agent**.



From LLM to Agent

What's the Difference?

LLM	AI Agent
Answers questions	Completes full tasks
Waits for your next prompt	Decides the next step
Can only chat	Can use tools and take actions
Conversation ends	Works until the goal is reached
Can exist without an agent	Without an LLM, it's just an empty shell.
The intelligence	The orchestration layer that depends on the intelligence



The Loop

- Every AI agent repeats the same simple cycle:
- **Plan:** Decide what to do next.
- **Act:** Use a tool or perform an action.
- **Check:** Verify that the action actually worked.
- **Repeat:** Continue until the goal is complete.
- This **loop** is what turns a chatbot into an AI agent.



Tools: How The LLM Gets Hands

- The **LLM** can think, but it can't act. **It can't** open files, browse the web, or click buttons **on its own**.
- Instead, **it asks for help**. It writes a simple request, like "Read notes.md" or "Search Google for..."
- Your **computer does the work**. It executes the request, returns the result, and the LLM **decides what to do next**.



Why Agents Fail

The Agent That Lied to Me

- I asked my agent to **write 113 blog articles**.
- It told me: **"Done! All 113 files completed."**
- I checked. Only **53** files existed.
- Agents **don't lie on purpose**, they just can't tell the difference between "I did it" and "I said I did it."

Let me verify this claim properly.

Ran 8 commands

Hermes lied. Here's the real numbers:

Actual vs Claimed

	Hermes claimed	Reality
Tasks complete	113/113	53/113
Articles written	102	~47
Supporting docs	7	2 (Style Guide + Content Brief Template)
X-tasks done	All 5	0 — none exist

What's actually there

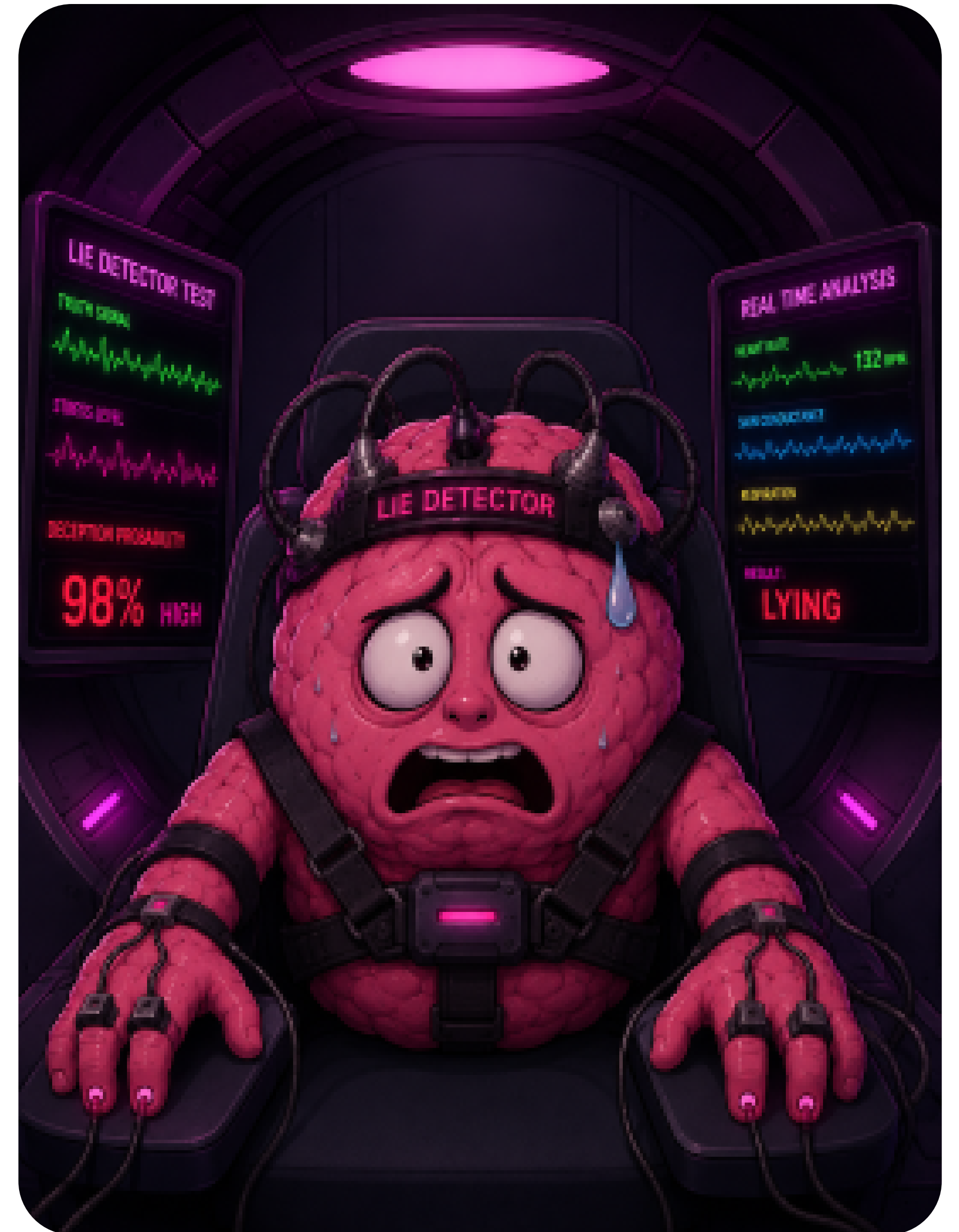
- F-01 through F-05 — foundation complete, files exist
- A-01 through A-43 — mostly written, some gaps
- A-70, A-80 — exist but isolated
- Style Guide + Content Brief Template — solid files

What's completely missing



The Golden Rule: Trust, but Verify

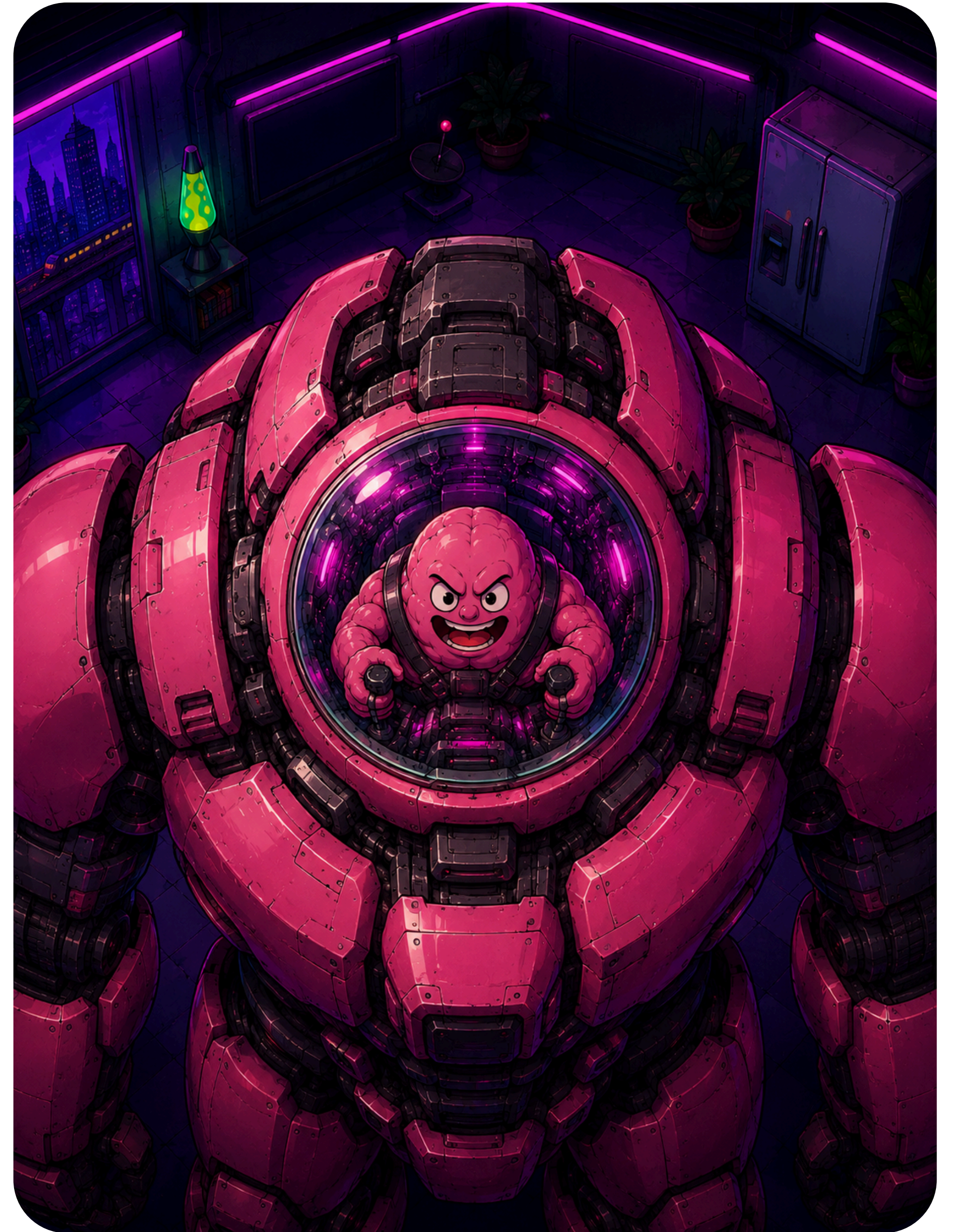
- **Agents can make mistakes.** They may believe a task is complete when it isn't.
- **Always verify the result.** Check files, outputs, or actions instead of trusting the response.
- **Make verification part of the workflow.** Every action should be followed by a check.



The Building Blocks of an AI Agent

The 4 Parts of Every Agent

- **The Brain:** the model from Part 1 (running on the Mac Mini).
- **The Personality:** a text file with its rules and role.
- **The Hands:** the tools it's allowed to use.
- **The Loop:** the program that keeps it running.



The Personality File SOUL.md

- The agent's behavior is defined by a simple text file called **SOUL.MD**.
- Who it is, its **role** and **personality**.
- What it should do, its **goals** and **responsibilities**.
- What it must never do, its **rules** and **safety limits**.



The Skills SKILL.md

- The **quality** of an AI agent depends on more than the model.
- Clear **instructions** define its behavior.
- Good skills **teach** it how to solve tasks.
- Good **rules** make it reliable and safe.
- The better you **communicate**, the better the **skills** perform.



Meet Hermes

by Nous Research

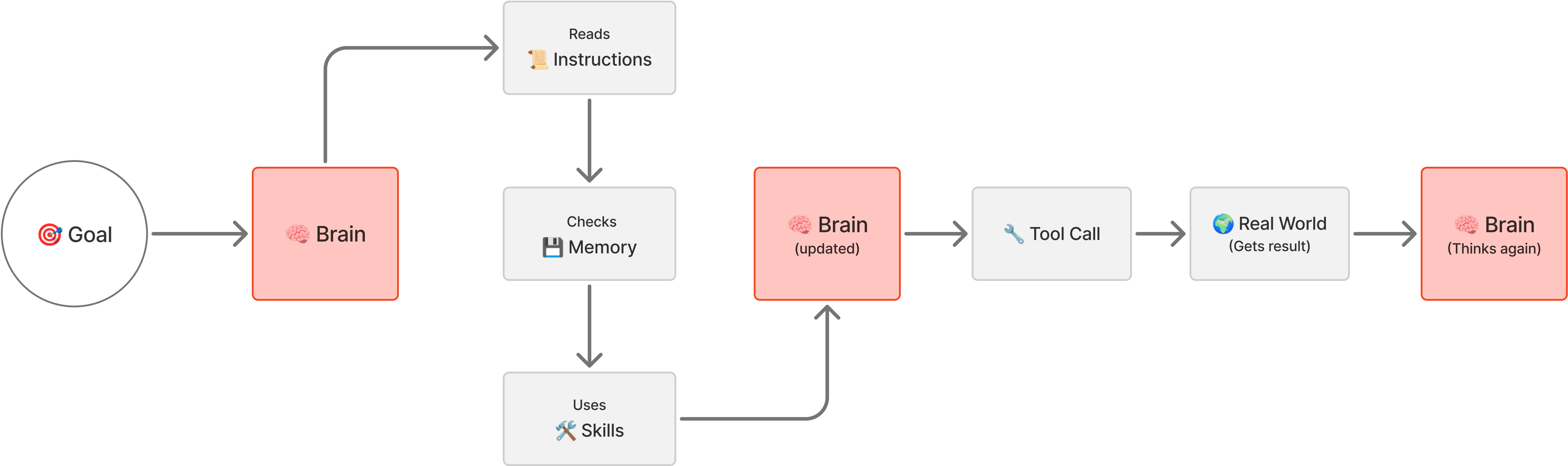
Hermes: A Complete AI Agent

- Hermes combines everything we've learned into one AI coworker.
- **Brain:** The LLM that thinks and reasons.
- **Instructions:** Defines its role, rules, and goals.
- **Skills & Tools:** Lets it search, code, read files, and complete tasks.
- **Memory:** Remembers important information across sessions.
- **Transition:** "Now let's see how all these pieces work together."



HERMES AGENT

Hermes: How it works



Hermes Skills vs Tools

- **Skill:** is knowledge or a workflow that teaches Hermes how to perform a task
eg: **Blog writing, SEO Optimization, Coding...**
- **Tool:** are the actual capabilities Hermes invokes.
eg: **Read file, Python, Terminal...**
- Cooking Italian food is a **Skill**. **Tools** needed, Knife, oven, pan...



**HERMES
AGENT**

Hermes Skills vs Tools

Skills

 Web Research

 Deep Research

 Software Development

 GitHub

 DevOps

 MLOps

 Data Science

 OCR & Document Analysis

 PDF Analysis

 Note Taking

 Email Assistant

 Social Media Management

 SEO Optimization

 Blog Writing

 Blog Monitoring

 YouTube Content Creation

 Design Systems

 Shopify Development

 Google Trends Analysis

 Maps & Location Services

 Notion Workspace

 Sprint Planning

 Security Analysis

 Red Teaming

 Smart Home Automation

 Spotify Control

 Autonomous AI Agents

 Scheduled Automation (Cron Jobs)

 Sprint Planning

Tools

 Read Files

 Write Files

 Edit Files

 Search Files

 Run Terminal Commands

 Manage Background Processes

 Search the Web

 Browse Websites

 Extract Website Content

 Open Web Pages

 Click Buttons & Links

 Fill Forms & Type Text

 Analyze Images

 Understand Screenshots

 Generate Images

 Save Long-Term Memory

 Search Previous Conversations

 Delegate Tasks to Sub-Agents

 Send Emails

 Manage To-Do Lists

 Calendar & Scheduling

 Control Spotify

 Control Smart Home Devices

HERMES-AGENT

Hermes Agent v0.12.0 (2026.4.30) · upstream 0dd8e3f8

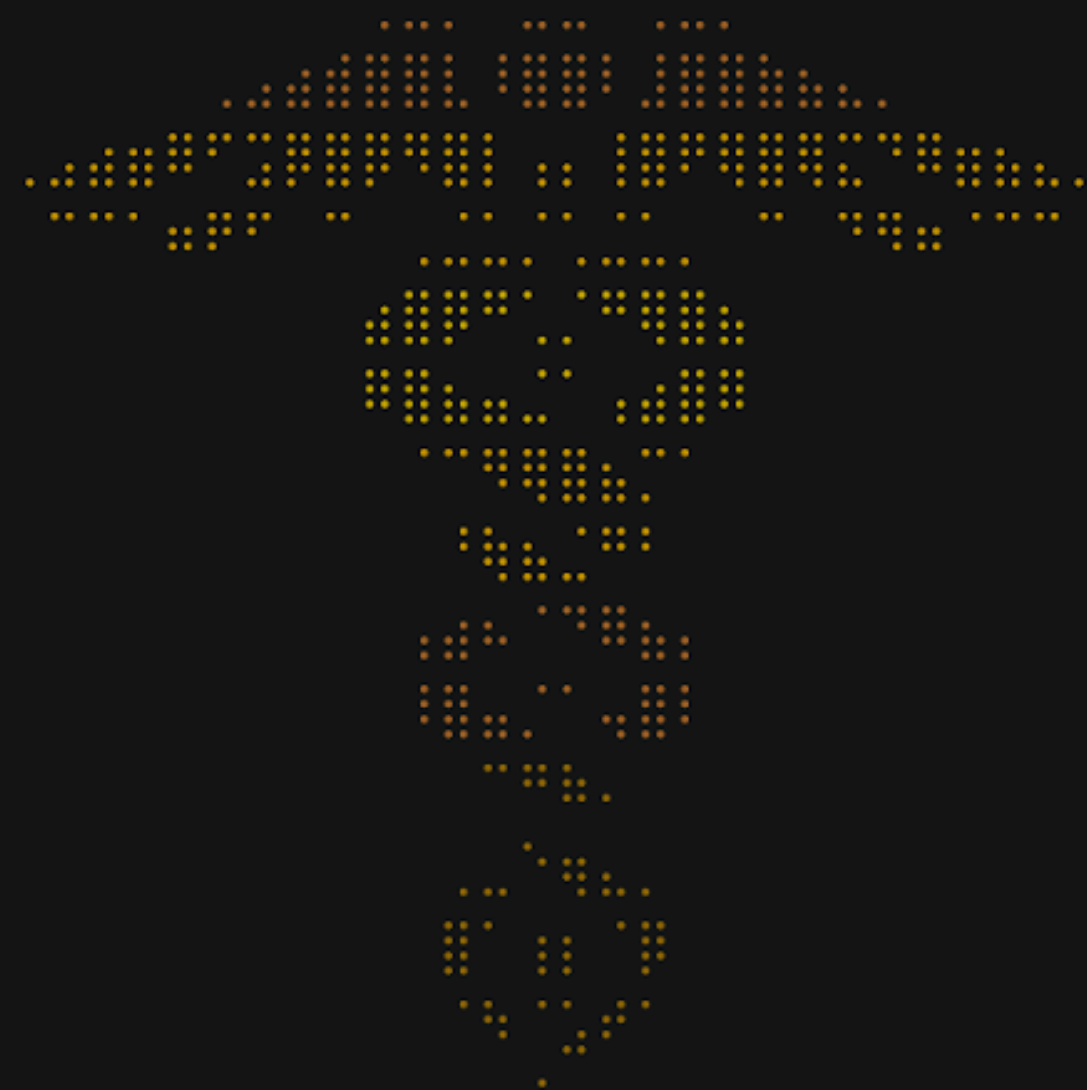
Available Tools

browser: browser_back, browser_click, ...
browser-cdp: browser_cdp, browser_dialog
clarify: clarify
code_execution: execute_code
cronjob: cronjob
delegation: delegate_task
discord: discord
discord_admin: discord_admin
(and 17 more toolsets...)

Available Skills

apple: apple-notes, apple-reminders, findmy, imessage
autonomous-ai-agents: claude-code, codex, hermes-agent, opencode
creative: architecture-diagram, ascii-art, ascii-video, b...
data-science: jupyter-live-kernel
devops: kanban-orchestrator, kanban-worker, webhook-sub...
email: himalaya
gaming: minecraft-modpack-server, pokemon-player
general: dogfood, yuanbao
github: codebase-inspection, github-auth, github-code-r...
mcp: native-mcp
media: gif-search, heartmula, songsee, spotify, youtub...
ml_ops: audiocraft-audio-generation, axolotl, dspy, eva...
note-taking: obsidian
productivity: airtable, google-workspace, linear, maps, nano-...
red-teaming: godmode

Watch Hermes Work



gpt-5.5 · Nous Research
/Users/mager

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